Expt. No. 7 Date 30/04/26

Page No. 49

Aim of the experiment

Design and Test 2-bit Comparator

Objective

To verify 2 bit comparator using universal logic gate.

Apparatus Required

SL NO	Name of component	Specification	Quantity
1.	Trainer kit (analog and digital)	XPO-ANADIG1	1
2.	Bread board	—	1
3.	AND Gate (IC-74LS11)	Triple 3-input AND Gate	1
4.	OR Gate (IC 74LS32)	Quad 2 input OR Gate	1
5.	NOT Gate (IC 74LS 04)	Hex inverter	1
6.	XOR Gate (IC 74LS 86)	Quad 2 input XOR Gate	1
7	Connecting wires	—	As per Required

Teacher's Signature _____

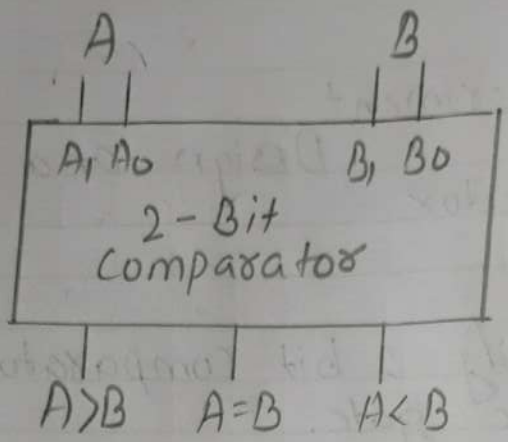


Fig 7.1: Block Diagram of 2 Bit Comparator

Expt. No. 7 Date 30/09/26

Page No. 50

Theory

A 2-bit Comparator is a combinational logic circuit used to compare two bit binary number $A = A_1A_0$ and $B = B_1B_0$. It determines whether A is less than, equal to, or greater than B.

The circuit has two 2-bit inputs and three output : $A < B$, $A = B$, and $A > B$. At a time only one output becomes High depending on the comparison between the two binary number

Truth Table :-

Table 7.1 :-

Input				Output		
A_1	A_0	B_1	B_0	$A<B$	$A=B$	$A>B$
0	0	0	0	0	1	0
0	0	0	1	1	0	0
0	0	1	0	1	0	0
0	0	1	1	1	0	0
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	1	0	0
0	1	1	1	1	0	0
1	0	0	0	0	0	1
1	0	0	1	0	0	1
1	0	1	0	0	1	0

Teacher's Signature

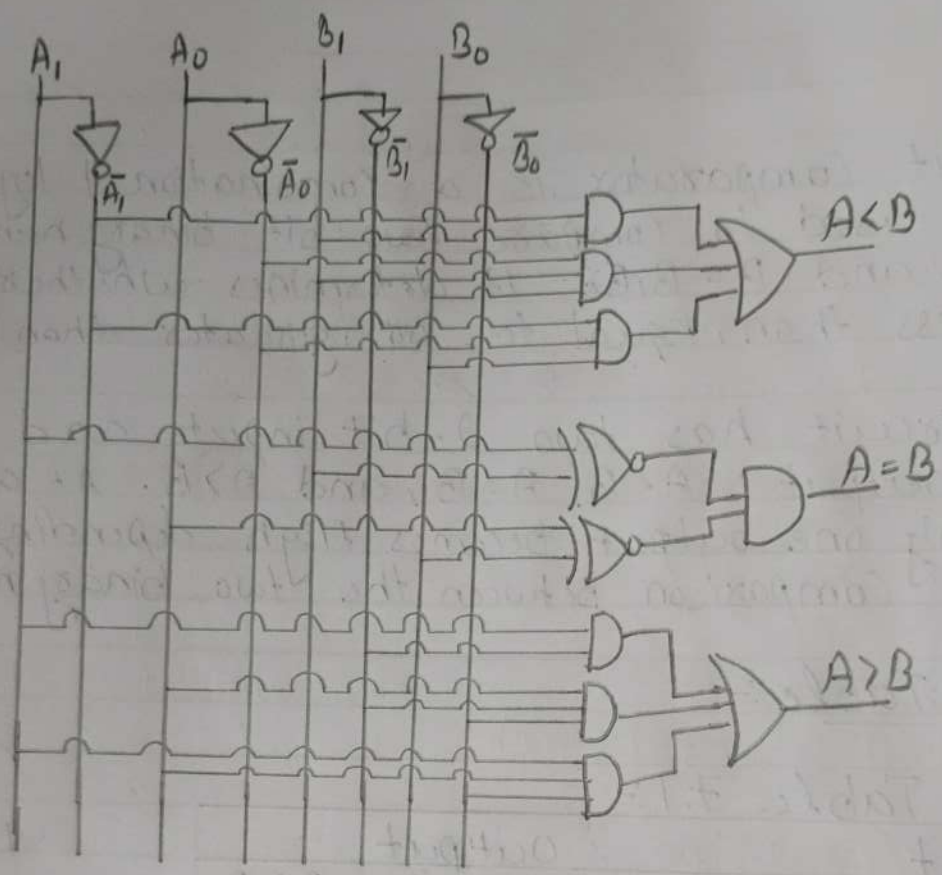


Fig 7.2: Logic circuit digram of 2-Bit comparator

Expt. No. 7 Date 30/09/26

Page No. 51

Table 7.1

Input				Output		
A_1	A_0	B_1	B_0	$A<B$	$A=B$	$A>B$
1	0	1	1	1	0	0
1	1	0	0	0	0	1
1	1	0	1	0	0	1
1	1	1	0	0	0	1
1	1	1	1	0	1	0

The Boolean expressions for the outputs are

1. For $A<B$

$$(A<B) = \Sigma m(1,2,3,6,7,11)$$

A_1A_0 \ B_1B_0	$\bar{B}_1\bar{B}_0$ 00	$\bar{B}_1B_0$ 01	B_1B_0 11	$B_1\bar{B}_0$ 10
$\bar{A}_1\bar{A}_0$ 00	0	1 2	1 3	1 2
$\bar{A}_1A_0$ 01	4	5	1 7	1 6
A_1A_0 11	12	13	15	14
$\bar{A}_1A_0$ 10	8	9	1 11	10

$$(A<B) = \bar{A}_1B_1 + \bar{A}_1\bar{A}_0B_0 + \bar{A}_1B_1B_0$$

2. For $A=B$

$$(A=B) = \Sigma m(0,5,10,15)$$

Teacher's Signature

Expt. No. 7 Date 30/09/26

Page No. 52

A_1A_0 \ B_1B_0	$\bar{B}_1\bar{B}_0$ 00	$\bar{B}_1B_0$ 01	B_1B_0 11	$B_1\bar{B}_0$ 10
$\bar{A}_1\bar{A}_0$ 00	(1) 0	1	3	2
$\bar{A}_1A_0$ 01	4	(1) 5	7	6
A_1A_0 11	12	13	(1) 15	14
$A_1\bar{A}_0$ 10	8	9	11	(1) 10

$$(A = B) = (A_1 \odot B_1)(A_0 \odot B_0)$$

3. For $A > B$

$$(A > B) = \Sigma m(4, 8, 9, 12, 13, 14)$$

A_1A_0 \ B_1B_0	$\bar{B}_1\bar{B}_0$ 00	$\bar{B}_1B_0$ 01	B_1B_0 11	$B_1\bar{B}_0$ 10
$\bar{A}_1\bar{A}_0$ 00	0	1	3	2
$\bar{A}_1A_0$ 01	(1) 4	5	7	6
A_1A_0 11	(1) 12	(1) 13	15	(1) 14
$A_1\bar{A}_0$ 10	(1) 8	(1) 9	11	10

$$(A > B) = A_1\bar{B}_1 + A_1A_0\bar{B}_0 + A_0\bar{B}_1\bar{B}_0$$

Procedure

1. Identify the required ICs and collect the necessary connecting wires and digital trainer kit.

Teacher's Signature

Expt. No. 7 Date 30/09/26

Page No. 53

7. Study the Boolean expression for the outputs $A<B$, $A=B$, and $A>B$ obtained using K-map simplification

3. Implement the simplified Boolean expression using the required universal logic gates

4. Connect the inputs A_1, A_0, B_1 and B_0 to the respective logic gates circuits.

5. Apply all possible combination of input value to the circuit

6. Observe the output corresponding to $A<B$, $A=B$, and $A>B$ for each input combination

7. Record the observed outputs in the truth table.

8. Compare the observed outputs with the theoretical truth table to verify the correct operation of the 2-bit Comparator

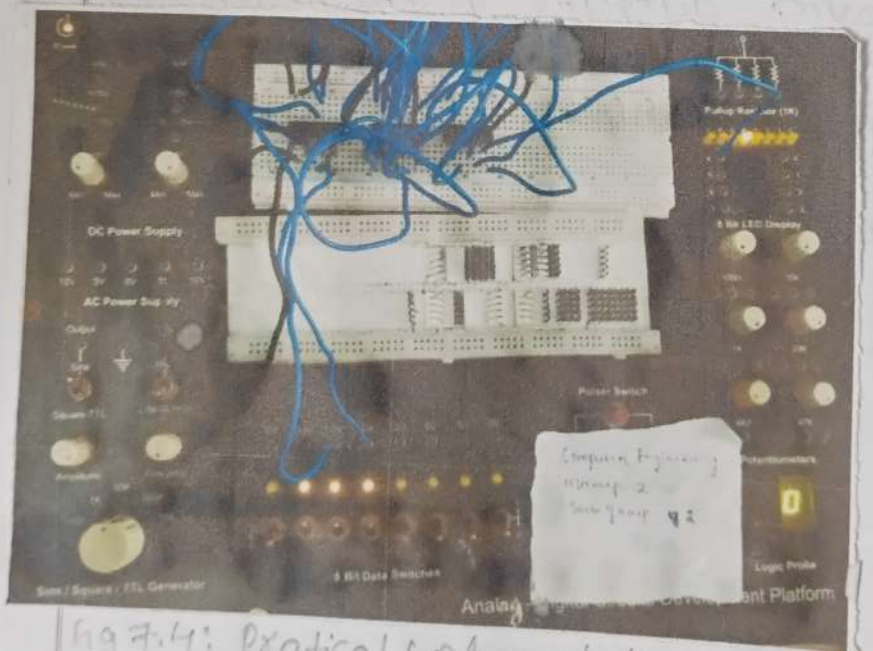
Teacher's Signature



Input				Output
$A_1(D_7)$	$A_0(D_6)$	$B_1(D_5)$	$B_0(D_4)$	$Y = A > B$ (4)
ON	OFF	OFF	OFF	Glowing

(ON = 1, OFF = 0)

Fig 7.3: Pratical Implementation of A > B for 2-Bit Comparator



Input				Output
A_1 (D_7)	A_0 (D_6)	B_1 (D_5)	B_0 (D_4)	$Y = A < B$ (L1)
OFF	ON	ON	ON	Glowing

(ON = 1, OFF = 0)

Fig 7.4: Pratical implimentation of A < B for 2-bit Comparato

Expt. No. 7 Date 30/09/26

Page No. 54

Observation Table

1. For A>B

Observation Table 1:-

Input				Output	
A_1 (D_7)	A_0 (D_6)	B_1 (D_5)	B_0 (D_4)	Y= A>B (4)	Remark
OFF	ON	OFF	OFF	Glowing	
ON	OFF	OFF	OFF	Glowing	The practical output matches the theoritical output
ON	OFF	OFF	ON	Glowing	
ON	ON	OFF	OFF	Glowing	
ON	ON	OFF	ON	Glowing	
ON	ON	ON	OFF	Glowing	

(ON=1, OFF=0)

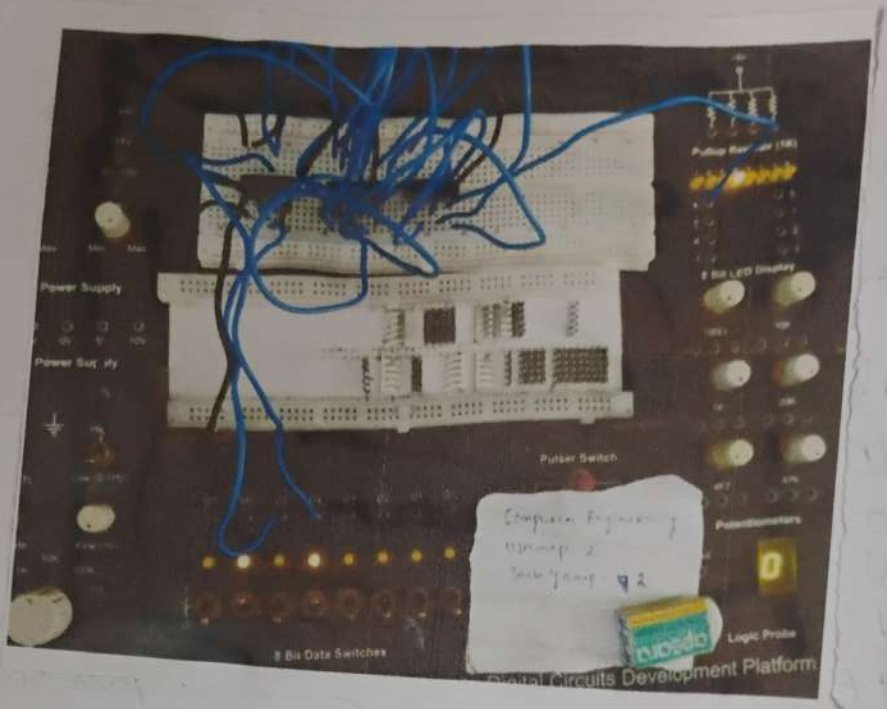
2. For A<B

observation Table 2:-

Input				Output	
A_1 (D_7)	A_0 (D_6)	B_1 (D_5)	B_0 (D_4)	Y=A<B (4)	Remark
OFF	OFF	OFF	ON	Glowing	
OFF	OFF	ON	OFF	Glowing	The pratical output matches the theoritical output
OFF	OFF	ON	ON	Glowing	
OFF	ON	ON	OFF	Glowing	
OFF	ON	ON	ON	Glowing	
ON	OFF	ON	ON	Glowing	

(ON=1, OFF=0)

Teacher's Signature



Input				Output
A_1 (D_7)	A_0 (D_6)	B_1 (D_5)	B_0 (D_4)	$Y=(A=B)$ (L1)
OFF	ON	OFF	ON	Glowing

(ON=1, OFF=0)

Fig 7.5: Pratical Implementation of A=B for 2-bit Comparator.

Expt. No. _

3. A

Expt. No. 7 Date 30/09/26

Page No. 55

3. For A=B

Observation Table 3:-

Input				Output	Remark
$A_1(D_7)$	$A_0(D_6)$	$B_1(D_5)$	$B_0(D_4)$	Y=(A=B)(4)	
OFF	OFF	OFF	OFF	Glowing	The pratical
OFF	ON	OFF	ON	Glowing	output matches
ON	OFF	ON	OFF	Glowing	theortical
ON	ON	ON	ON	Glowing	Output

(ON=1 , OFF=0)

Conclusion

The 2-bit Comparator has been designed and tested using universal logic gates The boolean expressions for A<B, A=B, A>B have been implemented and the correct operation of the circuit has been verified by comparing the observed outputs with the theoretical truth Table for all possible input combination

Name- Sambit Kumar Jena
Regd no- 25110283
Branch- Computer Engineering
Group - A2
Sub group - 2

Teacher's Signature